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Research Interests: Affective Computing, Large Language Models, Machine Learning

EDUCATION

BSMS Computer Science | De La Salle University, Manila, Philippines

2022 – Present

- Consistent Dean's Lister | CGPA: 3.5 / 4.0
- Relevant Coursework: Large Language Models, Machine Learning, Affective Computing and Multimodal Interactions, Design and Analysis of Algorithms, Advanced Software Engineering, Advanced Database Systems, Statistical Modeling and Simulation

EXPERIENCE

Nara Institute of Science and Technology (NAIST), HAI Lab

Jul. 2026 – Aug. 2026

Research Intern | JST Y-Tec Program

- Conducting research on applying Large Language Models (LLMs) to hydrology and Disaster Risk Reduction and Management (DRRM) at the Human-AI Interaction (HAI) Laboratory, as a selected participant in the JST Y-Tec exchange program.

AILRIC 2026 | De La Salle University

Jun. 2026

Reviewer | AI in Language Research International Conference, Student Research Workshop (SRW)

- Reviewed submitted papers for the AILRIC 2026 Student Research Workshop, evaluating research quality and providing feedback to emerging researchers in AI and language technology.

VISON Technologies Corporation

Aug. 2025 – Dec. 2025

Intern | Computer Vision Module Developer

- Optimized License Plate Recognition (LPR) modules by implementing Python, OpenCV, and OCR technologies, resulting in an 80% increase in character detection accuracy and confidence scores.
- Engineered robust image preprocessing pipelines to mitigate environmental noise and lighting variability, significantly improving OCR reliability for vehicle capture.

TE3D House | De La Salle University

Nov. 2024 – Present

Researcher | Technology, Education, Entertainment, Empathy, Design (TE3D) House

- Developed an emotion recognition pipeline that achieved 89% accuracy on the DAiSEE dataset using ensemble learning techniques.
- Processed and labeled over 9,000 frames of video data to train a multi-modal affective computing model.

Philippine Computing Science Congress | Computing Society of the Philippines

2025 – 2026

Editor | PCSC Conference Proceedings

- Systematized the compilation and formatting of accepted research papers using L^AT_EX for the PCSC 2025 and PCSC 2026 (26th PCSC, Davao City) proceedings, ensuring adherence to rigorous academic publishing standards and metadata consistency.
- Facilitated technical review workflows in collaboration with CSP board members to resolve document compilation errors and finalize manuscripts for publication.

PAPERS

- Sityar, L. A., & Azcarraga, J. (2026). *When More is Less: A Sensitivity Analysis of Geometric Landmarks, Facial Action Units, and Label Binarization for Affective State Recognition*. Presented at the 21st International Conference on Persuasive Technology (PERSUASIVE 2026), Hakodate, Japan.
- Carandang, M. R., Sityar, L. A., Lopez, D. J., & Azcarraga, J. (2026). *A Comparative Study of Quantum and Classical Machine Learning for Academic Emotion Recognition*. Accepted at the 2026 IEEE Region 10 Conference (TENCON 2026), Bali, Indonesia.
- Sityar, L. A., & Guillermo, J. (2025). *Design and Comparison of Agentic and Prompt-Engineered LLM Judges for English-Filipino Translation Evaluation*. De La Salle University.
- Sityar, L. A., & Carandang, M. R. (2025). *Design and Implementation of a Distributed Web Scraper for Link Extraction*. De La Salle University.
- Sityar, L. A., Alvarez, Y. E., Nuñez, J. C., & Villaver, R. M. (2024). *Predicting Emotions based on Text Input*. De La Salle University.
- Sityar, L. A., Garganera, A. J., Carandang, M. R., & Nogales, N. R. (2024). *Transaction Management on Steam Game Data: A Distributed Database Approach*. De La Salle University.
- Sityar, L. A., Garganera, A. J., & del Rosario, J. J. (2024). *Case Study on Multi-Tape Turing Machines*. De La Salle University.

PROJECTS

English-Filipino Translation Evaluation via Agentic LLM-as-a-Judge (*Natural Language Processing*)

- Developed and compared a structured Prompt-Engineered Judge and a multi-step Agentic Judge using a fine-tuned LLaMA-3-8B (4-bit quantized) model.
- Engineered a modular agentic framework featuring a short-term memory store and specialized linguistic tools, including an Idiom Checker and Word Connotation Checker.
- Conducted comparative analysis using Spearman's rank correlation, finding that agentic workflows provide superior qualitative explainability and pragmatic error detection.

Distributed Web Scraper for Link Extraction (*Distributed Systems*)

- Engineered a high-performance Master-Worker distributed architecture using Pyro5 RMI to facilitate Remote Method Invocation across multiple nodes.
- Overcame the Python Global Interpreter Lock (GIL) by implementing true process-level concurrency, achieving a throughput of 3,951 pages in 10 minutes.

Recallify: Spaced-Repetition Learning Platform (*Technical Lead & Architect*)

- Architected the end-to-end system logic, implementing a custom Spaced-Repetition System (SRS) algorithm to optimize memory retention.
- Led technical mentorship for the development team, conducting code reviews to standardize implementation and ensure architectural integrity.
- Established the project's DevOps infrastructure, including a Git-based CI/CD pipeline and Jest testing suite to ensure high-performance deployment.

Kiku: Primed Listening (*Browser Extension & Language Learning Tool*)

- Developed and published an open-source Chrome extension for YouTube and Netflix that automates "Primed Listening" by programmatically pausing playback after dialogue.
- Engineered real-time subtitle detection and playback control logic to facilitate language immersion, maintaining a 5-star rating on the Chrome Web Store.
- Implemented a privacy-focused, local-only architecture with customizable hotkeys and dynamic UI elements to optimize user learning workflows.

LEADERSHIP EXPERIENCE

QuERST Summer Virtual Tutorial (SVT) | De La Salle University

Apr. 2026 – May 2026

Student Volunteer Tutor / Mathematics

- Tutored an incoming Grade 5 student in daily one-on-one sessions over four weeks, reinforcing Grades 2-4 prerequisite math skills for QuERST's SDG 4 (Quality Education) initiative.

Society of Proactive Role Models INspiring Total Development | De La Salle University

Aug. 2024 – Present
Associate Vice President / Creatives Committee

- Collaborated with the Executive Board to translate organizational goals into creative strategies while managing multiple high-pressure deadlines concurrently.

SERVIR under the Office of the Associate Dean in Academics | De La Salle University

Oct. 2023 – Present
Student Volunteer / Head - Creatives and Communications Committee

- Spearheaded marketing for AMPLIFY (150+ delegates) and leads a creatives team supporting 10+ annual academic events.

Student Discipline Formation Unit Paragons | De La Salle University

Oct. 2023 – Present

Student Volunteer / Executive - Creatives Committee

- Project-managed Advocacy Run initiatives across three external committees and drove a 50% increase in social media reach.

Computer Studies Government | De La Salle University

Jul. 2023 – Jul. 2024

Student Volunteer / Executive - Integrated Marketing Communications

- Managed the communications pipeline and visual assets for the 1,000+ member CCS student body.

TECHNICAL SKILLS & INTERESTS

- **Languages:** English, Filipino, Japanese (Intermediate)
- **Technical:** Python, Java, SQL, JavaScript, C, C++, Prolog, Go
- **Machine Learning Technologies:** LLMs, Agentic Frameworks, RAG, Prompt Engineering, PyTorch, Hugging Face Transformers, OpenCV, Scikit-learn, Pandas, NumPy, Sentiment Analysis
- **Platforms:** Google Colab, Anaconda, Git, Github, Visual Studio Code, Hugging Face, Proxmox, Figma, Adobe Illustrator
- **Interests:** Language Learning, Graphic Design, Boxing, Running